

ELECTROCHEMISTRY

Introduction

Electrochemistry refers to the conversion of chemical energy to electrical energy and vice versa. It is basically the study of Production of electricity from energy released during spontaneous reaction and use of electrical energy to bring about non-spontaneous chemical transformation.

A spontaneous chemical reaction is a reaction which happens on its own and releases free energy. This reaction produces electric energy from chemical reaction. For example, burning of coal, rusting of iron, melting of ice, etc.

On the other hand non-spontaneous reaction occurs by providing an external source like electricity. For example, Hydrolysis of water.

Electrochemistry is used for the following purposes;

- Production of metals like sodium hydroxide, chlorine, fluorine and many other chemicals.
- It is also used for purification of metals.
- The process is used in batteries as well as fuel cells which converts the chemical energy into electrical energy and is used in several instruments and devices.
- This process is used in electroplating.
- The reactions carried out using the process of electrochemistry are energy effective and less polluting.

Electrochemical cell:

The cell that converts the chemical energy liberated as a result of spontaneous redox reaction to electrical energy is called a galvanic cell or voltaic cell. E.g., Daniell cell.

- Daniell cell has an electrical potential of 1.1 V.
- If an external opposite potential is applied in the galvanic cell and increased slowly, we find that the reaction continues to take place till the opposing voltage reached the value 1.1V. When, the reaction stops altogether and no current flows through the cell. Any further increase in the external potential again starts the reaction but in opposite direction. It now functions as an electrolytic cell.

Thus, when $E_{\text{ext}} = 1.1\text{V}$, no flow of electrons and current and no chemical reaction.

When $E_{\text{ext}} > 1.1\text{V}$, electrons flow from Cu to Zn and current flow flows from Zn to Cu.

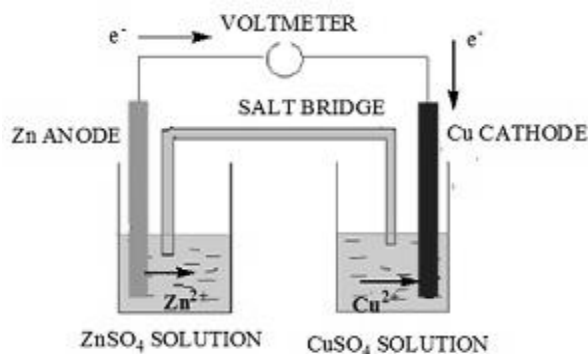
Thus, Zn deposited at zinc electrode and copper dissolves at copper electrode.

When $E_{\text{ext}} < 1.1\text{V}$, electrons flow from Zn to Cu and current flow flows from Cu to Zn.

Thus, Cu deposited at copper electrode and Zn dissolves at Zn electrode.

The setup for Daniell cell is as follows;

- In a beaker a plate of zinc is dipped in a solution of zinc sulfate (ZnSO_4). In another beaker a plate of copper is dipped in a solution of copper (II) sulfate in another container. These plates of metal are called the *electrodes* of the cell. These electrodes behave as terminal to hold the electrons.
- The two electrodes are connected via wire.
- A salt bridge is placed between the two beakers. This provides a path for the movement of ions from one beaker to the other in order to maintain electrical neutrality.
- Zinc electrode gets oxidized and hence releases electrons that flow through the wire towards the copper electrode.
- The copper (II) sulfate solution releases copper ions Cu^{2+} .



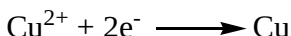
At the anode:

Oxidation: loss of electrons.



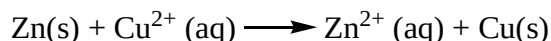
At cathode:

Reduction: gain of electrons.



Zinc atoms being more reactive have a greater tendency to lose electrons than that of copper. The electrons in this cell moves from zinc anode to copper cathode through the wire connecting the two electrodes in the external circuit

- A bulb placed within this circuit will glow and a voltmeter connected within this circuit will show deflection.
- The net reaction of this cell is the sum of two half-cell reactions (also called cell reaction).



Thus, the deduction half reaction occurs on the copper electrode while the oxidation half reaction occurs on the zinc electrode. These two portions of the cell are called Half-cells or redox couple. The copper electrode may be called the reduction half-cell and the zinc electrode, the oxidation half-cell.

Salt Bridge:

In a Daniel cell a salt bridge is placed between the two beakers containing a solution of zinc sulfate (ZnSO_4) and a solution of copper (II) sulfate respectively.

Salt bridge is an inverted U-tube like structure, filled with concentrated solution of an inert electrolyte with a mixture prepared by mixing agar-agar or gelatin. The electrolyte does not undergo any sort of chemical reaction nor does it reacts with concentrated solution in the two half cells. Most of the time salts like KCl , KNO_3 , NH_4NO_3 are used as electrolyte.

Uses of salt bridge:

- It completes the electrical circuit as it connects the electrolytes in the two half cells.
- It avoids the chances of diffusion of solutions between the half cells.
- It maintains electrical neutrality by providing a path for the movement of ions.

Representation of an electrochemical cell:

The following conversions are used in representing an electrochemical cell:

- A galvanic cell is represented by writing the anode (where oxidation occurs) on the left hand side and cathode (where reduction occurs) on the right hand side.

- The anode of the cell is represented by writing metal or solid phase first and then the electrolyte while the metal and the cation are separated by a vertical line. For further information, the concentration of the electrolyte is also mentioned within bracket after the cation.
- The salt bridge which separates the two half cells is indicated by two vertical lines.

Representation of some common cell:

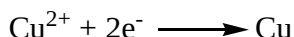
1. Daniell cell (Zn-CuSO₄ cell):

The reactions occurring in the cell are-

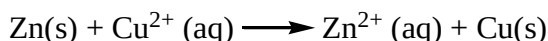
Oxidation half reaction;



Reduction half reaction;



Overall cell reaction;



The cell may be represented by:



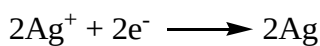
2. Cu-AgNO₃ cell:

The reactions occurring in the cell are-

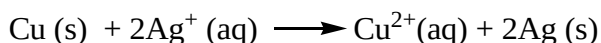
Oxidation half reaction;



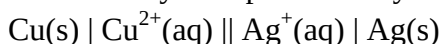
Reduction half reaction;



Overall cell reaction;



The cell may be represented by:



Electrode potential:

A potential difference develops between the electrode and the electrolyte which is called electrode potential. The electrode potential may be of two types:

1. Oxidation potential: The tendency of an electrode to lose electrons or to get oxidised is called its oxidation potential.
2. Reduction potential: the tendency of an electrode to gain electrons or to get reduced is called its reduction potential.

The half cell reactions are always written as reduction half cell and their potentials are represented as reduction potentials.

The electrode potential depends upon;

- (i) The nature of the metal and its ions,
- (ii) Concentration of the ions in solution, and
- (iii) Temperature

e.m.f. of the cell or cell potential:

The potential difference between two electrodes of the cell when either no or negligible current is allowed to flow in the circuit is called e.m.f. of the cell or cell potential. Thus,

$$\text{e.m.f.} = \text{reduction potential of cathode} - \text{reduction potential of anode}$$

$$\text{or, } E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$\text{or, } E_{\text{cell}} = E_{\text{right}} - E_{\text{left}}$$

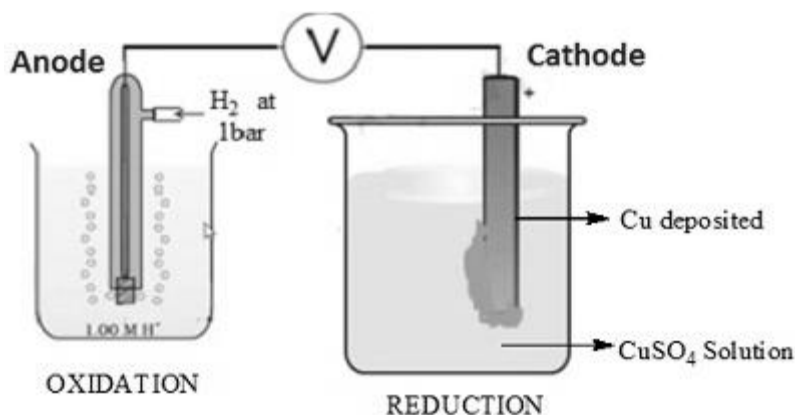
The e.m.f. of the cell is measured with the help of potentiometer. E.g., the e.m.f. of daniell cell of 1M at 298K is 1.10 volts.

Standard electrode potential:

When the concentration of all the species involved in a half-cell is unity (1M) and if the temperature and pressure is in standard state (at 298K and 1bar pressure) than the electrode potential is known as standard electrode potential.

Measurement of standard electrode potential of a cell:

- The electrode is connected to a **standard hydrogen electrode (SHE)** to constitute a cell.
- The electrode forming the negative terminal of the cell is allotted negative value of electrode potential whereas the electrode forming the positive terminal of the cell is allotted a positive value of electrode potential.
- The potential difference developed between the two terminals is measured using a potentiometer.
- The direction of the flow of electric current in the external circuit is identified using a galvanometer.
- This enables us to identify the positive and the negative terminal of the set up as the current flows from positive terminal to negative terminal.
- This in turn will help us mark the anode and the cathode electrodes due to the fact that the electrons will flow from anode to cathode.



$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

• In case of Daniel cell

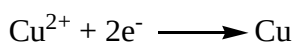
At the anode:

Oxidation: loss of electrons.

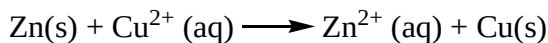


At cathode:

Reduction: gain of electrons.



The net reaction of this cell is the sum of two half-cell reactions (also called cell reaction).

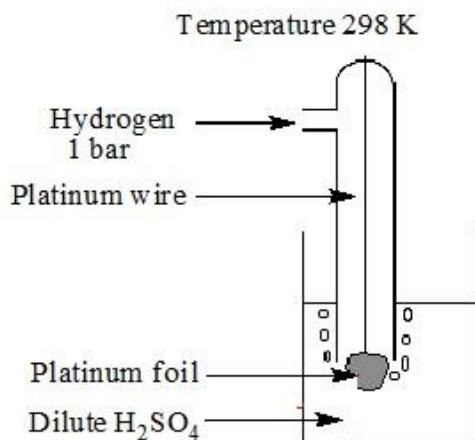
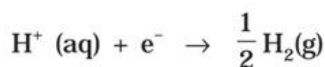


$$\begin{aligned} \text{e.m.f. of the cell} &= E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} \\ &= 0.34 \text{ V} - (-0.76) \text{ V} = 1.10 \text{ V} \end{aligned}$$

(Measured Emf of Cu is 0.34 V and that of Zn is 0.76 V).

Standard hydrogen electrode

It consists of a platinum electrode coated with a layer of platinum black. The electrode is immersed in an acidic solution and the pure hydrogen gas is bubbled through it.



The concentration of the reduced form and the oxidized form of hydrogen is sustained at unity with following conditions:

- Pressure of hydrogen gas = 1 bar
- Concentration of hydrogen ion in the solution = 1 molar

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$E_{\text{cell}} = E_{\text{cathode}} - 0 = E_{\text{cathode}}$$

- The measured e.m.f. of the cell:

$\text{Pt} | \text{H}_2 (1 \text{ bar}) | \text{H}^+ (1\text{M}) || \text{Cu}^{2+} (1\text{M}) | \text{Cu}$ is 0.34 V.

The positive value of the standard electrode potential signifies the easy reduction of Cu^{2+} ions than H^+ ions.

- The measured e.m.f. of the cell

$\text{Pt} | \text{H}_2 (1 \text{ bar}) | \text{H}^+ (1\text{M}) || \text{Zn}^{2+} (1\text{M}) | \text{Zn}$ is -0.76 V.

The negative value of the standard electrode potential signifies that the hydrogen ions oxidize the zinc (or it can be said that zinc can reduce hydrogen ions).

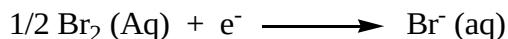
- An electrode with standard electrode potential greater than zero is stable in its reduced form compared to hydrogen gas.
- Whereas an electrode with negative standard electrode potential is less stable in its reduced form compared to hydrogen gas.

- This decreases the standard electrode potential which in turn decreases the oxidizing power of the specific electrode on the left and increases the reducing power of the electrode to the right of the reaction.

Other standard electrode is

Bromine electrode: $\text{Pt(s)}|\text{Br}_2(\text{aq})|\text{Br}^-(\text{aq})$

With half reaction;



Electrochemical series:

The arrangement of elements in order of increasing electrode potential values is called electrochemical series. The electrochemical series, also called activity series, of same typical electrodes is given in the table:

Half-reaction	E° (V)	Half-reaction	E° (V)
$\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$	2.87	$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$	0.40
$\text{Ag}^{2+} + \text{e}^- \rightarrow \text{Ag}^+$	1.99	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	0.34
$\text{Co}^{3+} + \text{e}^- \rightarrow \text{Co}^{2+}$	1.82	$\text{Hg}_2\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Hg} + 2\text{Cl}^-$	0.27
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$	1.78	$\text{AgCl} + \text{e}^- \rightarrow \text{Ag} + \text{Cl}^-$	0.22
$\text{Ce}^{4+} + \text{e}^- \rightarrow \text{Ce}^{3+}$	1.70	$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{SO}_3 + \text{H}_2\text{O}$	0.20
$\text{PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^- \rightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$	1.69	$\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$	0.16
$\text{MnO}_4^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{MnO}_2 + 2\text{H}_2\text{O}$	1.68	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00
$\text{IO}_4^- + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{IO}_3^- + \text{H}_2\text{O}$	1.60	$\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}$	-0.036
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51	$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$	-0.13
$\text{Au}^{3+} + 3\text{e}^- \rightarrow \text{Au}$	1.50	$\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn}$	-0.14
$\text{PbO}_2 + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.46	$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$	-0.23
$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	1.36	$\text{PbSO}_4 + 2\text{e}^- \rightarrow \text{Pb} + \text{SO}_4^{2-}$	-0.35
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.33	$\text{Cd}^{2+} + 2\text{e}^- \rightarrow \text{Cd}$	-0.40
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	1.23	$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.44
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.21	$\text{Cr}^{3+} + \text{e}^- \rightarrow \text{Cr}^{2+}$	-0.50
$\text{IO}_3^- + 6\text{H}^+ + 5\text{e}^- \rightarrow \frac{1}{2}\text{I}_2 + 3\text{H}_2\text{O}$	1.20	$\text{Cr}^{3+} + 3\text{e}^- \rightarrow \text{Cr}$	-0.73
$\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$	1.09	$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76
$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightarrow \text{VO}^{2+} + \text{H}_2\text{O}$	1.00	$2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$	-0.83
$\text{AuCl}_4^- + 3\text{e}^- \rightarrow \text{Au} + 4\text{Cl}^-$	0.99	$\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$	-1.18
$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO} + 2\text{H}_2\text{O}$	0.96	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	-1.66
$\text{ClO}_2 + \text{e}^- \rightarrow \text{ClO}_2^-$	0.954	$\text{H}_2 + 2\text{e}^- \rightarrow 2\text{H}^-$	-2.23
$2\text{Hg}^{2+} + 2\text{e}^- \rightarrow \text{Hg}_2^{2+}$	0.91	$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$	-2.37
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	0.80	$\text{La}^{3+} + 3\text{e}^- \rightarrow \text{La}$	-2.37
$\text{Hg}_2^{2+} + 2\text{e}^- \rightarrow 2\text{Hg}$	0.80	$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	-2.71
$\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$	0.77	$\text{Ca}^{2+} + 2\text{e}^- \rightarrow \text{Ca}$	-2.76
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}_2$	0.68	$\text{Ba}^{2+} + 2\text{e}^- \rightarrow \text{Ba}$	-2.90
$\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_4^{2-}$	0.56	$\text{K}^+ + \text{e}^- \rightarrow \text{K}$	-2.92
$\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$	0.54	$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$	-3.05
$\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$	0.52		

Application of the electrochemical series:

- **Prediction of relative strengths of oxidizing and reducing agents:**

With the help of electrochemical series, we can predict the relative oxidizing or reducing strengths of substances. In general, higher the value of standard electrode potential stronger is the oxidizing agent and lower the value of standard electrode potential stronger is the reducing agent. E.g.,

- It can be seen that the standard electrode potential of fluorine is highest in the table indicating that the fluorine gas (F_2) has maximum tendency to get reduced to fluoride ions (F^-) and therefore, fluorine gas is the strongest oxidizing agent and fluoride ion is weakest reducing agent.
- Lithium has the lowest electrode potential in the table indicating that lithium has maximum tendency to get oxidized and hence lithium metal is the most powerful reducing agent or lithium ion is the weakest oxidizing agent in aqueous solution.

- **Calculation of e.m.f. of the cell:**

The e.m.f. of the cell which is the difference between the reduction potential of the cathode and anode determined by the following equation

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

- **Prediction of feasibility of the reaction:**

The electrochemical series helps in finding out whether the reaction is feasible or not in general, a redox reaction is feasible only if the species which has higher reduction potential is reduced i.e., accepts the electrons and the species which has lower electrode potential are oxidized i.e., loses the electron. In other words species to release electrons must have lower electrode potential as compared to the species which is to accept electrons.

- **To predict whether a metal can liberate hydrogen from acid or not:**

Electrochemical series can also be used to predict whether a metal can liberate hydrogen from the acid or not. In general, only those metals can liberate hydrogen from the acid which has negative values of reduction potentials. e.g., the positive value of standard electrode potential in the first case indicates that Cu^{2+} ion gets reduced more easily than H^+ ions. The reverse process cannot occur, that is, hydrogen ions cannot oxidize Cu (or alternatively we can say that hydrogen gas can reduce copper ion.) under the standard conditions. Thus Cu does not dissolve in HCl.

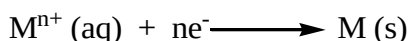
In nitric acid it is oxidized by nitrate ion not by hydrogen ion.

The negative value of standard electrode potential of zinc indicates that hydrogen ions can oxidize zinc (or zinc can reduce hydrogen ions).

Nernst Equation

The Nernst Equation empowers the assurance of cell potential under non-standard conditions and relates the measured cell potential to the reaction quotient and permits the exact measurement of equilibrium constants.

Let us consider an electrochemical reaction of the following type:



Nernst equation for this can be written as follows:

$$E_{\text{M}^{n+}/\text{M}} = E_{\text{M}^{n+}/\text{M}}^{\circ} - \frac{RT}{nF} \ln \frac{[\text{M}]}{[\text{M}^{n+}]}$$

Here, $E_{\text{M}^{n+}/\text{M}}$ = electrode potential (at any concentration)

$E_{\text{M}^{n+}/\text{M}}^{\circ}$ = standard electrode potential of metal M

R = gas constant ($8.314 \text{ JK}^{-1} \text{ mol}^{-1}$)

T = temperature in Kelvin

[M] = molar concentration of metal (or solid phase)

$[\text{M}^{n+}]$ = concentration of M^{n+}

F = Faraday constant (96487 C mol^{-1})

But the concentration of the solid phase [M(s)] is taken as unity and thus, Nernst equation can be written as;

$$E_{\text{M}^{n+}/\text{M}} = E_{\text{M}^{n+}/\text{M}}^{\circ} - \frac{RT}{nF} \ln \frac{1}{[\text{M}^{n+}]}$$

In case of Daniell cell Nernst equation is as follows:

For cathode:

$$E_{Cu^{2+}/Cu} = E_{Cu^{2+}/Cu}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[Cu^{2+}]}$$

For anode:

$$E_{Zn^{2+}/Zn} = E_{Zn^{2+}/Zn}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[Zn^{2+}]}$$

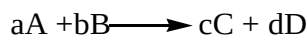
The cell potential, $E_{cell} = E_{Cu^{2+}/Cu} - E_{Zn^{2+}/Zn}$

$$\begin{aligned} &= E_{Cu^{2+}/Cu}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[Cu^{2+}]} - E_{Zn^{2+}/Zn}^{\circ} + \frac{RT}{2F} \ln \frac{1}{[Zn^{2+}]} \\ &= E_{Cu^{2+}/Cu}^{\circ} - E_{Zn^{2+}/Zn}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[Cu^{2+}]} + \frac{RT}{2F} \ln \frac{1}{[Zn^{2+}]} \\ &E_{cell} = E_{cell}^{\circ} - \frac{RT}{2F} \ln \frac{[Zn^{2+}]}{[Cu^{2+}]} \end{aligned}$$

by converting the natural logarithm to the base 10 and substituting the values of R, F and $T = 298K$. it reduces to

$$E_{cell} = E_{cell}^{\circ} - \frac{0.059}{2} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

For a general electrochemical reaction of the type:

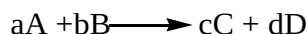


Nernst equation can be written as:

$$E_{cell} = E_{cell}^{\circ} - \frac{RT}{nF} \ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Equilibrium constant from Nernst equation:

For a general electrochemical reaction of the type:



Nernst equation can be written as:

$$E_{cell} = E_{cell}^{\circ} - \frac{2.303RT}{nF} \log \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

But at equilibrium, $E_{cell} = 0$ and $\frac{[C]^c [D]^d}{[A]^a [B]^b} = K_c$

Thus, the above equation can be written as

$$\begin{aligned} E_{cell} = 0 &= E_{cell}^{\circ} - \frac{2.303RT}{nF} \log K_c \\ E_{cell}^{\circ} &= \frac{2.303RT}{nF} \log K_c \end{aligned}$$

This is the relationship between equilibrium constant of the reaction and standard potential of the cell in which that reaction takes place.

Electrochemical Cell and Gibbs Energy of the Reaction:

Electrical work done (1 second) = Electrical potential X Total charge passed

Passing the charges reversibly through the galvanic cell results in maximum work.

Reversible work done by galvanic cell = Decrease in Gibbs energy

Let E = Emf of the cell

nF = Amount of charge passed

$\Delta_r G$ = Gibbs energy of the reaction

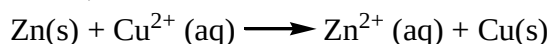
$$\Delta_r G = -nFE_{cell}$$

If the concentration of all the reacting species is unity, then $E_{cell} = E_{cell}^0$

And we have,

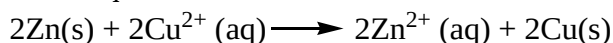
$$\Delta_r G^0 = -nFE_{cell}^0$$

For the reaction,



$$\Delta_r G = -2FE_{cell}$$

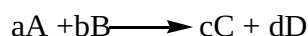
But when the equation becomes



$$\Delta_r G = -4FE_{cell}$$

Relation between Gibbs free energy and equilibrium constant:

For a general electrochemical reaction of the type:



We know that, $\Delta_r G^0 = -nFE_{cell}^0$

$$\text{And, } E_{cell}^0 = \frac{2.303RT}{nF} \log K_c$$

Comparing the above equations, we get,

$$\begin{aligned} \Delta_r G^0 &= -nF \frac{2.303RT}{nF} \log K_c \\ \Delta_r G^0 &= -2.303RT \log K_c \end{aligned}$$

Conductance of electrolytic solution:

It is necessary to define a few terms before we consider the subject of conductance of electricity through electrolytic solutions;

Electrical resistance: it measures the obstruction to the flow of current and it is denoted by the symbol ' R '.

- It is measured in a unit called ohm (Ω) with the help of a Wheatstone bridge
- According to Ohm's law, the electrical resistance of any conductor is:
 - (i) Directly proportional to its length, l
 - (ii) Inversely proportional to its area of cross section, A . i.e.,

$$R \propto \frac{1}{A}$$

$$R = \rho \frac{1}{A}$$

In the equation the constant of proportionality denoted by 'ρ' (rho) is called **resistivity**. It is measured in units called ohm metre (S.I. system) or ohm cm.

When length = 1m.

And Cross sectional area = 1m²

Then resistance becomes the resistivity.

Thus, resistivity for a substance is defined as its resistance when it is 1 metre long and its area of cross section is 1 m².

$$1 \Omega \text{ m} = 100 \Omega \text{ cm}$$

$$1 \Omega \text{ cm} = 0.01 \Omega \text{ m}$$

Electrical conductance

The inverse of resistance is known as conductance and is denoted by 'G'.

$$G = \frac{1}{R} = \frac{A}{\rho l} = \kappa \frac{A}{l}$$

It is measured in a unit called Siemens represented by the symbol 'S'. It is equal to ohm⁻¹ also known as mho or Ω⁻¹.

The inverse of resistivity is termed as conductivity or specific conductance. It is represented by the symbol κ (Greek, kappa). It is measured in a unit called Sm⁻¹.

When length = 1m and Cross sectional area = 1m², then conductivity becomes the conductance.

Thus, conductivity or specific conductance of a substance is defined as its conductance when it is 1 m long and its area of cross section is 1 m².

$$1 \text{ s cm}^{-1} = 100 \text{ s m}^{-1}$$

Matters can be classified into conductors, insulators and semiconductors depending on the magnitude of their conductivity.

Conductors:

Solids with conductivities ranging between 10⁴ to 10⁷ ohm⁻¹m⁻¹ are called conductors. Metals have conductivities in the order of 10⁷ ohm⁻¹m⁻¹ is good conductors. For example, Iron, Copper, Aluminum etc.

Insulators:

Solids with very low conductivities ranging between 10⁻²⁰ to 10⁻¹⁰ ohm⁻¹m⁻¹ are called insulator. For example Wood, plastic, cloth, glass.

Semiconductors:

Solids with conductivities in the intermediate range from 10⁻⁶ to 10⁴ ohm⁻¹m⁻¹ are called semiconductor. For example Gallium, Germanium, Silicon.

Superconductor:

Superconductors are the substance which has zero resistivity or infinite conductivity. Earlier, only metals and their alloys at very low temperatures (0-15K) were known to behave as superconductors, but

nowadays a number of ceramic materials and mixed oxides are also known to show superconductivity at temperature as high as 150K.

Conductors are broadly classified into two types:

Metallic or electronic conductance:

These are metallic substances which allow the electricity to pass through them without undergoing any chemical change. Electrical conductance through metals is called metallic conductance or electronic conductance. E.g., copper, silver etc. The electronic conductance depends on:

1. The nature and structure of the metal
2. The number of valence electrons per atom
3. Temperature (it decreases with increase of temperature).

Electrolytic or ionic conductance:

These are substances which allow the electricity to pass through them in their molten state or in the form of their aqueous solutions and undergo chemical decomposition. The flow of current through an electrolytic solution is called electrolytic conductance. In this type of conduction, charge is carried by ions. Therefore, it is also called ionic conductance.

Non-electrolyte: the substance, which do not conduct electricity either in their molten state or through their aqueous solutions are called non-electrolyte. E.g., sugar, glucose, urea, etc.

Difference between metallic and electrolytic conduction:

Metallic conduction	Electrolytic conduction
1. Metallic conduction is carried by the movement of electrons.	1. Electrolytic conduction is carried by the movement of ions.
2. It involves no change in chemical properties of the conductor.	2. It involves the decomposition of the electrolyte as a result of the chemical reaction.
3. It does not involve the transfer of any matter.	3. It involves the transfer of matter as ions.
4. Metallic conduction decreases with increase in temperature.	4. Electrolytic conduction increase with increase in temperature.

Classification of electrolytes:

Electrolytes are broadly classified into two types:

1. Strong electrolytes: the electrolytes which are almost completely dissociated into ions in solution are called strong electrolytes. E.g., NaCl, KCl, NaOH etc.
2. Weak electrolytes: the electrolytes which do not ionize completely in solution are called weak electrolytes. E.g., CH_3COOH , H_3BO_3 , HCN etc.

Factors affecting electrical conductivity of electrolytic solution:

The conductivity of electrolytic solution depends upon the following factors:

1. **Nature of electrolytes:** The conductance of an electrolytic depends upon the number of ions present in the solution. The greater the number of ions in the solution, greater is the conductance.
2. **Nature of the solvent:** Electrolytes ionize more in polar solution. Therefore, greater the polarity of the solvent, larger is the ionization and hence greater is the conductance.
3. **Size of the ions produced and their solution:** If the ions are strongly solvated, their effective size will increase and hence their conductance will decrease.
4. **Concentration of the electrolytic solution:** Higher the concentration of the solution, less is the conductance.
5. **Temperature:** with increase in temperature, the conductivity of an electrolyte increase due to decrease in interactions between ions.

Measurement of conductivity of a solution of an electrolyte

The conductance of is the reciprocal of the resistance, therefore, the experimental determination of the conductance of a solution involves the measurement of its resistance.

However, for measuring the resistance of an ionic solution, we face two difficulties:

1. For electrolytic solution, direct current cannot be used because it causes electrolysis of the solution. To overcome this problem, alternating current is provided by using a suitable electronic circuit is used.
2. A solution of unknown resistance cannot be connected to the bridge like metallic wire or other solid conductor. This problem is solved by using a specially designed vessel called conductivity cell for taking solution.

Basically it consist of two platinum electrodes coated with platinum block. These have the area of cross section equal to 'A' and are separated by distance 'l'.

In the formula

$$R = \rho \frac{l}{A} = \frac{l}{\kappa A}$$

Here, l/A is called **cell constant** denoted by the symbol, G^* . The cell constant depends on the distance between the electrodes (l) and their area of cross-section (A).

It is determined by measuring the resistance of the cell containing a solution of known conductivity. The cell constant G^* , is than given by the equation:

$$G^* = \frac{l}{A} = R \cdot \kappa$$

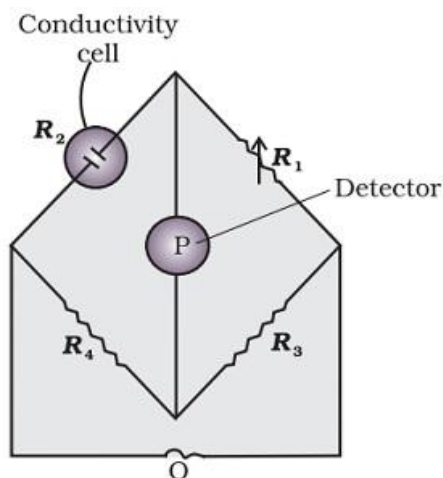
Arrangement for measurement of Resistance

The arrangement consists of two resistances R_3 and R_4 . There is a variable resistance R_1 and a conductivity cell with unknown resistance R_2 . The Wheatstone bridge is provided with an oscillator O that acts as source of a.c. power. The arrangement has a suitable detector P. The Wheatstone bridge is balanced when there is no flow of current through the detector. Under this condition:

$$\text{Unknown resistance} = R_2 = R_1 R_4 / R_3$$

After calculating the resistance the conductivity can be easily calculated using the formula:

$$\kappa = G^* / R$$



Molar conductivity

Molar conductivity is defined as the conducting power of all the ions produced by dissolving one mole of an electrolyte in solution. It is denoted by the symbol ' Λ_m '. It is related to the conductivity of the solution by the following equation:

$$\text{molar conductivity} = \Lambda_m = \frac{\kappa}{C}$$

In the above equation, if κ is expressed in Sm^{-1} and the concentration, C in mol m^{-3} then the units of Λ_m are $\text{S m}^2 \text{mol}^{-1}$. It may be noted that:

$1 \text{ mol m}^{-3} = 1000(\text{L/m}^3) \times \text{molarity (mol/L)}$ and hence,

$$\Lambda_m(\text{S m}^2 \text{mol}^{-1}) = \frac{\kappa(\text{S m}^{-1})}{1000(\text{Lm}^{-3}) \times \text{molarity (mol L}^{-1})}$$

If we use S cm^{-1} as the units for κ and mol cm^{-3} , the units of concentration, then the units for Λ_m are S cm^2 . It can be calculated by using the equation:

$$\Lambda_m(\text{S cm}^2 \text{mol}^{-1}) = \frac{\kappa(\text{S cm}^{-1}) \times 1000(\text{cm}^3 \text{L}^{-1})}{\text{molarity (mol L}^{-1})}$$

Both types of units are used in literature and are related to each other by the equations:

$$\begin{aligned} 1 \text{ S m}^2 \text{mol}^{-1} &= 10^4 \text{ S cm}^2 \text{mol}^{-1} \\ \text{Or, } 1 \text{ S cm}^2 \text{mol}^{-1} &= 10^{-4} \text{ S m}^2 \text{mol}^{-1} \end{aligned}$$

Variation of Conductivity and Molar Conductivity with Concentration:

On dilution, Conductivity of both weak and strong electrolytes decreases with decrease in concentration while molar conductivity increases. This is because, as the number of ions per unit volume carrying the current in a solution decreases on dilution, the conductivity decreased.

On the other hand, the molar conductivity is the product of conductivity (κ) and volume (V) of the solution containing one mole of the electrolyte.

$$\Lambda = \kappa \times V$$

On dilution, conductivity decreases but volume containing one mole of an electrolyte increases. It has been found that the increase in volume on dilution is much more than the decrease in conductivity. As a result molar conductivity increases on dilution.

Molar conductivity at infinite dilution or limiting molar conductivity:

When the concentration approaches to zero, the molar conductivity is referred to as limiting molar conductivity. It is represented by the symbol Λ_m^0 .

Variation of Molar Conductivity with Concentration for strong electrolyte:

For strong electrolytes, Λ_m increases slowly with dilution and can be represented by the equation:

$$\Lambda_m = \Lambda_m^0 - A c^{1/2}$$

This equation is called Debye-Huckel equation.

Where, Λ_m = molar conductivity

Λ_m^0 = limiting molar conductivity

C = concentration of electrolyte

Λ = constant, for a given solvent and temperature it depends on the types of electrolyte i.e., the charges on the cation and anion produced on the dissociation of electrolyte in the solution.

The variation of molar conductivity with concentration can be studied by plotting Λ_m vs $c^{1/2}$. The slope for Λ_m vs $c^{1/2}$ is linear at low concentrations. The extrapolation of graph to zero concentration gives the value of limiting molar conductivity.

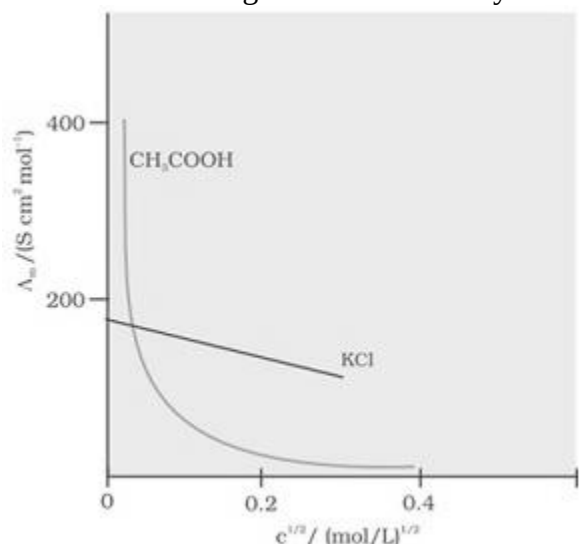


Fig. Molar conductivity Λ_m Vs. $c^{1/2}$ for acetic acid and potassium chloride

Variation of Molar Conductivity with Concentration for weak electrolyte:

The slope for Λ_m vs $c^{1/2}$ for weak electrolytes is not linear even at lower concentrations. So we cannot obtain limiting molar conductivity by extrapolation of the Λ_m vs $c^{1/2}$. But it can be obtained by indirect method based upon Kohlrausch law of independent migration of ion.

Kohlrausch law of independent migration of ions

The Kohlrausch law of independent migration of ions states that limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte.

Limiting equivalent molar conductivity of an electrolyte is the algebraic sum of limiting molar equivalent conductivities of its constituent ions.

Limiting Molar conductivities of Cations & Anions @ 298K

Ion	$\lambda^0/(\text{S cm}^2\text{mol}^{-1})$	Ion	$\lambda^0/(\text{S cm}^2\text{mol}^{-1})$
H ⁺	349.6	OH ⁻	199.1
Na ⁺	50.1	Cl ⁻	76.3
K ⁺	73.5	Br ⁻	78.1
Ca ²⁺	119.0	CH ₃ COO ⁻	40.9
Mg ²⁺	106.0	SO ₄ ²⁻	160.0

Mathematically,

$$\Lambda^0 = v_+ \lambda_+^0 + v_- \lambda_-^0$$

Where λ_+^0 = limiting equivalent conductivities of cation

λ_-^0 = limiting equivalent conductivities of anion.

v_+ and v_- are the number of cations and anions per formula unit of electrolyte

e.g., for NaCl, $\Lambda^0(\text{NaCl}) = \lambda^0(\text{Na}^+) + \lambda^0(\text{Cl}^-)$

Calculation of molar conductance at infinite dilution for weak electrolyte by using Kohlrausch's law:

Let us calculate the limiting molar conductivity for CH_3COOH . According to Kohlrausch's law,

$$\Lambda^0(\text{CH}_3\text{COOH}) = \lambda^0(\text{CH}_3\text{COO}^-) + \lambda^0(\text{H}^+)$$

This equation can be obtained by the knowledge of molar conductivity at infinite dilution for some strong electrolytes. For example, consider the strong electrolytes HCl , NaCl and CH_3COOH . From Kohlrausch's law,

$$\Lambda^0(\text{CH}_3\text{COONa}) = \lambda^0(\text{CH}_3\text{COO}^-) + \lambda^0(\text{Na}^+)$$

$$\Lambda^0(\text{HCl}) = \lambda^0(\text{H}^+) + \lambda^0(\text{Cl}^-)$$

$$\Lambda^0(\text{NaCl}) = \lambda^0(\text{Na}^+) + \lambda^0(\text{Cl}^-)$$

It is clear that,

$$\lambda^0(\text{CH}_3\text{COO}^-) + \lambda^0(\text{H}^+) = [\lambda^0(\text{CH}_3\text{COO}^-) + \lambda^0(\text{Na}^+)] + [\lambda^0(\text{H}^+) + \lambda^0(\text{Cl}^-)] - [\lambda^0(\text{Na}^+) + \lambda^0(\text{Cl}^-)]$$

$$\text{Or, } \Lambda^0(\text{CH}_3\text{COOH}) = \Lambda^0(\text{CH}_3\text{COONa}) + \Lambda^0(\text{HCl}) - \Lambda^0(\text{NaCl})$$

$$\text{Similarly, } \Lambda^0(\text{NH}_4\text{OH}) = \Lambda^0(\text{NH}_4\text{Cl}) + \Lambda^0(\text{NaOH}) - \Lambda^0(\text{NaCl})$$

Degree of dissociation and dissociation constant for weak electrolyte:

At any concentration c , if α is the degree of dissociation than it can be approximated to the ratio of molar conductivity at the concentration c to limiting molar conductivity. Thus,

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0}$$

But we know that for weak electrolyte like acetic acid,

$$K_a = \frac{c\alpha^2}{(1-\alpha)} = \frac{c\Lambda_m^0}{\left(1 - \frac{\Lambda_m}{\Lambda_m^0}\right)} = \frac{c\Lambda_m^0}{\Lambda_m^0(\Lambda_m^0 - \Lambda_m)}$$

Electrolytic cells and electrolysis:

Electrolysis is a process of decomposition of an electrolyte by the passage of electricity through its aqueous solution or molten state of electric current. The cells used to carry out electrolysis are electrolytic cells.

The process of electrolysis is carried out by taking the solution of electrolyte in a suitable vessel, called electrolytic tank. Two metallic rods or plates are suspended in the electrolytic solution. These are connected to the terminal of a battery with the help of metallic wires. These metallic rods or plates allow the passage of current and are called electrodes. The electrode connected to the positive terminal of battery is called anode while electrode to the negative terminal of the battery is called cathode.

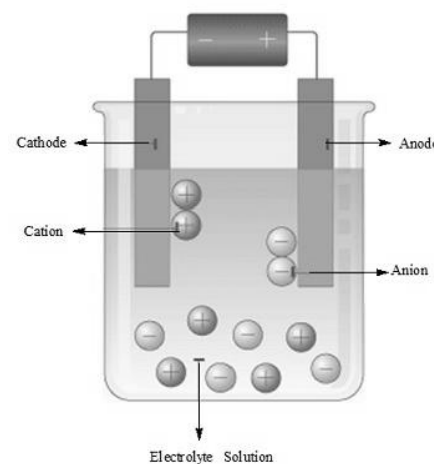


Fig. Electrolysis

Quantitative aspects of electrolysis:

Michael Faraday described the quantitative aspects of electrolysis and came forward with two laws of electrolysis:

Faraday's Laws of Electrolysis

First Law: the amount of any substance deposited or liberated at any electrode is directly proportional to the quantity of electricity passed through the electrolytes.

Thus, if w grams of the substance is deposited on passing Q coulombs of electricity, then

$$w \propto Q \quad \text{or} \quad w = zQ$$

Where z is a constant of proportionality and is called electrochemical equivalent. If a current of I amperes is passed for t seconds, then

$$Q = I \times t$$

So that, $Q = z \times I \times t$

Now, if $Q = 1$ coulomb

$I = 1$ ampere and $t = 1$ second,

Then $w = z$

Hence, electrochemical equivalent of a substance may be defined as the mass of the substance deposited when a current of one ampere is passed for one second i.e., a quantity of electricity equal to one coulomb is passed.

Second Law: The amounts of different substances deposited or liberated by the same quantity of electricity passing through the electrolytic solution are proportional to their chemical equivalent weights

Thus when same current is passed through two electrolytic solutions, containing two electrolytes A and B with equivalent weight E_A and E_B respectively connecting in a series and the weight of A and B deposited is W_A and W_B then according to Faraday's law,

$$\frac{W_A}{W_B} = \frac{E_A}{E_B}$$

The charge on one mole of electrons is called 1 **Faraday** and is represented by the symbol **F**.

$1F = 96485 \text{ C}$ or approximately 96500 C .

Products of Electrolysis

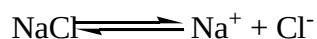
The product of electrolysis produced depends on the followings;

1. Nature of electrolytes (molten or aqueous)
2. Nature of electrodes
3. Electrode potential of electrolytes (in cathode reaction containing higher E^0 value is preferred and in anode reaction containing lower E^0 value is preferred)

Overvoltage: it may be noted that some electrochemical processes, though feasible, are so slow kinetically that at lower voltages, these do not seem to occur. Therefore, for occurrence of such reactions some extra potential or voltage is required than the theoretical value of their standard electrode potential. This extra voltage required is called over voltage.

Product of electrolysis:

Electrolysis of molten NaCl: the molten sodium chloride has Na^+ and Cl^- ions,

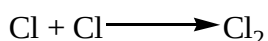
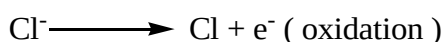


When electric current is passed through molten sodium chloride, the Cl^- ions are attracted towards anode and Na^+ ion moves towards the cathode. Thus,

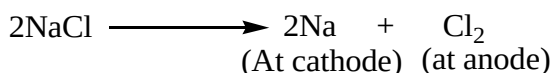
At cathode:



At anode:



Thus, the overall reaction is:



Electrolysis of aqueous sodium chloride: in water, sodium chloride ionizes as:

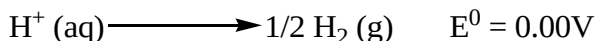
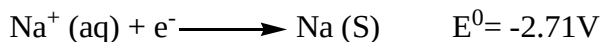


Water also dissociates into ions as:

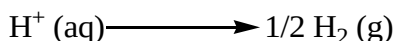


When electricity passed through the solution the following reactions are occur:

At cathode: there is a competition for reduction between the following reactions:



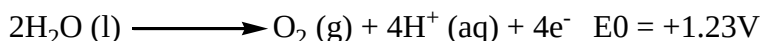
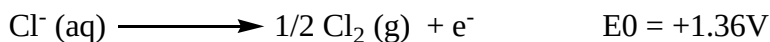
The reaction with higher value of E^0 is preferred and therefore, the reaction at the cathode is:



But H^+ ions are produced by the dissociation of water. Therefore, the net reaction at the cathode may be written as:

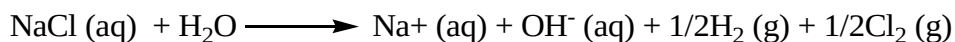


At anode: the following reactions are possible:

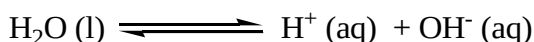


The standard reduction potential of water is slightly less. Therefore, it has slightly more chances of getting oxidized. However, in concentrated solution of NaCl , oxidation of chloride ion is preferred than water at anode and therefore Cl_2 gas is liberated. The unexpected result is due to the fact that sometimes higher voltage is required for electrolysis due to over voltage.

Thus the net reactions during the electrolysis of aqueous sodium chloride solution may be written as:

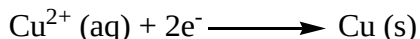


Electrolysis of aq. CuSO_4 by using Pt-electrode: CuSO_4 and H_2O ionise as:

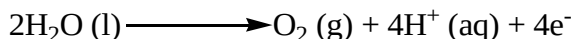


When current is passed, the following reactions are occurs at electrode:

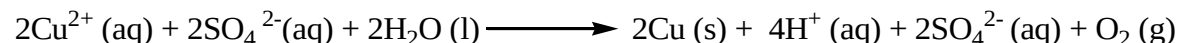
At cathode:



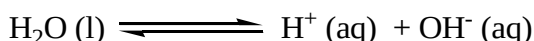
At anode:



The overall reaction is:

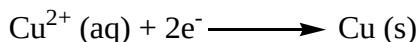


Electrolysis of aq. CuSO_4 by using Cu-electrode: CuSO_4 and H_2O ionise as:



When current is passed, the following reactions are occurs at electrode:

At cathode:



At anode: Unlike electrolysis of CuSO_4 using Pt-electrodes, no ions are liberated. On the other hand, anode itself undergoes loss of electrons to form Cu^{2+} ions which go into the solution.

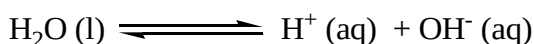
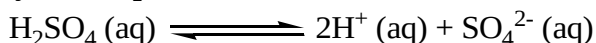


So, the net result of electrolysis is that copper is deposited at the cathode from the solution and equivalent amount of copper from the anode dissolves into solution as Cu^{2+} ions.

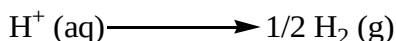
Q. Write the product of electrolysis of (i) Aq. AgNO_3 by using Pt-electrode, (ii) Aq. AgNO_3 by using Ag-electrode.

Ans: similar to electrolysis of CuSO_4 Solution.

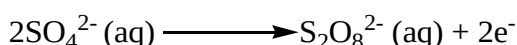
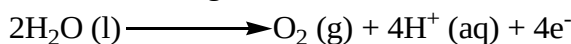
Electrolysis of $\text{aq. H}_2\text{SO}_4$: H_2SO_4 and H_2O ionizes as



At cathode: During the electrolysis of sulphuric acid, hydrogen is liberated at cathode:

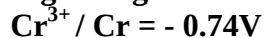
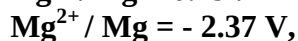
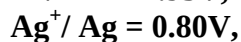
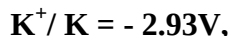


At anode: The following reactions occur at anode:



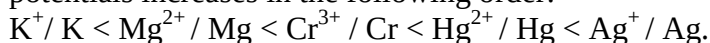
For dilute solution first reaction is preferred and O_2 liberates at anode. But at higher concentration of sulphuric acid, i.e., second reaction is preferred and solution of per oxodisulphuric acid ($\text{H}_2\text{S}_2\text{O}_8$) is obtained.

Problem: Given the standard electrode potentials,



Arrange these metals in their increasing order of reducing power.

Solution: Lower the reduction potential leads to higher reducing power. The given standard electrode potentials increases in the following order:



Hence, reducing power of the given metals increases in the following order:



Battery:

Battery is an arrangement of electrochemical cells that can be used as sources of direct electric current at a constant voltage.

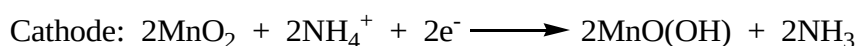
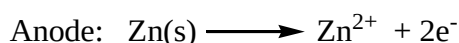
It is of two types:

1. Primary battery
2. Secondary battery

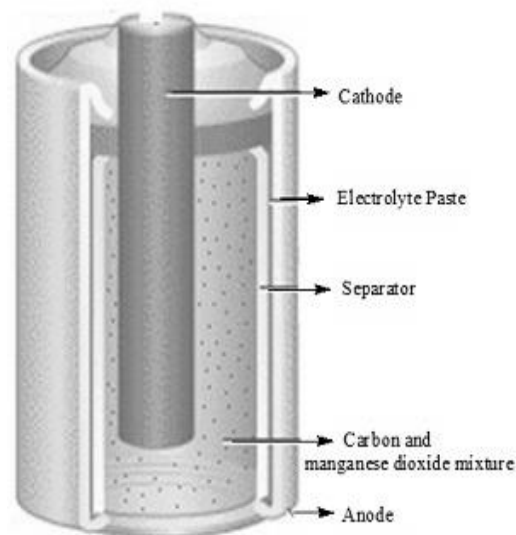
Primary battery:

In the primary batteries, the reaction occurs only once and after use over a period of time battery becomes dead and cannot be reused again. The most familiar example of this type is the dry cell (known as Leclanche cell after its discoverer).

Dry cell: The cell consists of a zinc container that also acts as anode and the cathode is a carbon (graphite) rod surrounded by powdered manganese dioxide and carbon (Fig.3.8). The space between the electrodes is filled by a moist paste of ammonium chloride (NH_4Cl) and zinc chloride (ZnCl_2). The electrode reactions are complex, but they can be written approximately as follows



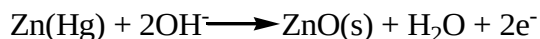
- At the cathode manganese is reduced from + 4 oxidation state to +3 oxidation state.
- As per the equation, Ammonia produced forms a complex with Zn^{2+} giving rise to $[\text{Zn}(\text{NH}_3)_4]^{2+}$.
- The cell has a potential of nearly 1.5 V.
- It is used commonly in our transistors and clocks.



Mercury cell: Mercury cell is suitable for low current devices like hearing aids, watches, etc.

Mercury cell consists of zinc – mercury-amalgam as the anode and a paste of HgO and carbon as the cathode. The electrolyte is a paste of KOH and ZnO . The electrode reactions for the cell are given below:

Anode:



Cathode:



The overall reaction is represented by

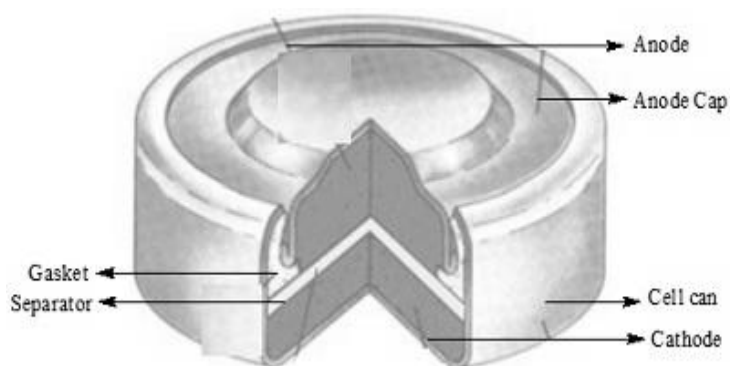
$$\text{Zn(Hg)} + \text{HgO(s)} \longrightarrow \text{ZnO(s)} + \text{Hg(l)}$$


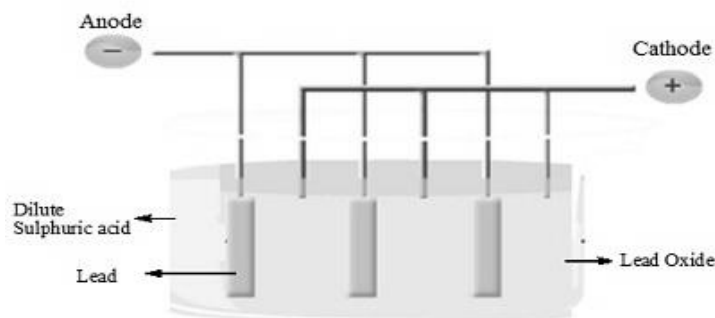
fig: Mercury cell

The cell potential is approximately 1.35 V that remains constant throughout its lifetime as the overall reaction does not involve any ion in solution whose concentration can change during its life time.

Secondary Battery

A secondary cell after use can be recharged by passing current through it in the opposite direction so that it can be used again. A good secondary cell can undergo a large number of discharging and charging cycles. The most important secondary cell is the lead storage battery.

Lead storage battery: It is commonly used in automobiles and invertors. It consists of a lead



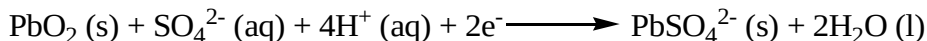
anode and a grid of lead packed with lead dioxide (PbO_2) as cathode. A 38% solution of sulphuric acid is used as an electrolyte.

The cell reactions when the battery is in use are given below:

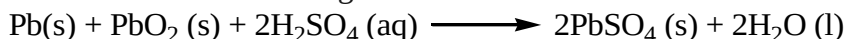
Anode:



Cathode:

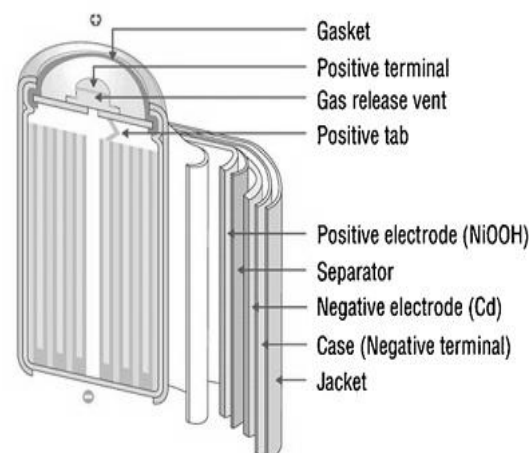


i.e., overall cell reaction consisting of cathode and anode reactions is:



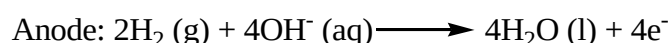
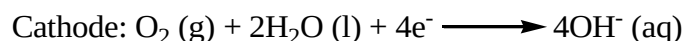
On charging the battery the reaction is reversed and $\text{PbSO}_4(\text{s})$ on anode and cathode is converted into Pb and PbO_2 , respectively.

Cadmium battery: Another important secondary cell is the nickel-cadmium cell which has longer life than the lead storage cell but more expensive to manufacture. We shall not go into details of working of the cell and the electrode reactions during charging and discharging. The overall reaction during discharge is:

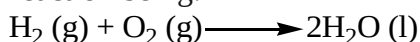


Fuel cells: Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, methanol, etc. directly into electrical energy are called fuel cells.

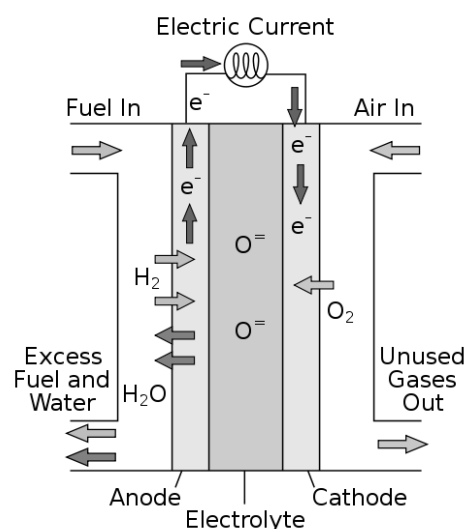
One of the most successful fuel cells uses the reaction of hydrogen with oxygen to form water. The cell was used for providing electrical power in the Apollo space programme. The water vapours produced during the reaction were condensed and added to the drinking water supply for the astronauts. In the cell, hydrogen and oxygen are bubbled through porous carbon electrodes into concentrated aqueous sodium hydroxide solution. Catalysts like finely divided platinum or palladium metal are incorporated into the electrodes for increasing the rate of electrode reactions. The electrode reactions are given below:



Overall reaction being:



The cell runs continuously as long as the reactants are supplied. Fuel cells produce electricity with an efficiency of about 70 % compared to thermal plants whose efficiency is about 40%. There has been tremendous progress in the development of new electrode materials, better catalysts and electrolytes for increasing the efficiency of fuel cells. These have been used in automobiles on an experimental basis. Fuel cells are pollution free and in view of their future importance, a variety of fuel cells have been fabricated and tried.



Corrosion: Corrosion is a process of deterioration of a metal as a result of its reaction with air or water (environment) surrounding it. The rusting of iron, tarnishing of silver, development of green coating on copper and bronze are some of the examples of corrosion.

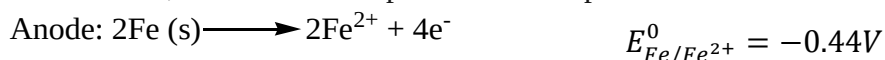
In case of iron, the corrosion is called rusting. Chemically, rusting is hydrated form of ferric oxide, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.

Factors which effect corrosion:

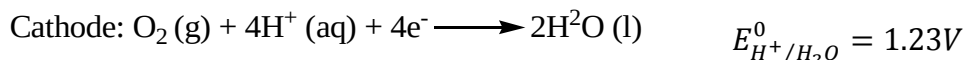
1. **Position of metals in e.m.f. series:** More the reactivity of metal, the more will be the possibility of the metal getting corroded.
2. **Presence of impurities in metals:** The presence of impurity help in setting up voltaic cells, which increases the speed of corrosion.
3. **Presence of electrolytes:** Presence of electrolytes in water also increases the rate of corrosion. E.g., corrosion of iron in sea water takes place to longer extent than distilled water.

Mechanism of rusting of iron:

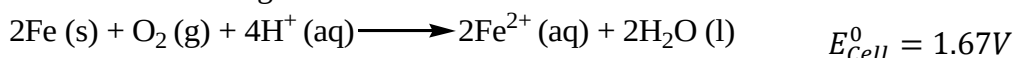
It may be considered essentially as an electrochemical phenomenon. At a particular spot of an object made of iron, oxidation takes place and that spot behaves as anode and we can write the reaction:



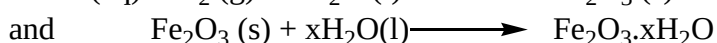
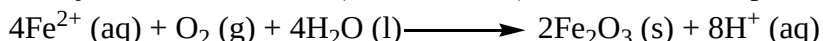
Electrons released at anodic spot move through the metal and go to another spot on the metal and reduce oxygen in presence of H^+ (which is believed to be available from H_2CO_3 formed due to dissolution of carbon dioxide from air into water. Hydrogen ion in water may also be available due to dissolution of other acidic oxides from the atmosphere). This spot behaves as cathode with the reaction:



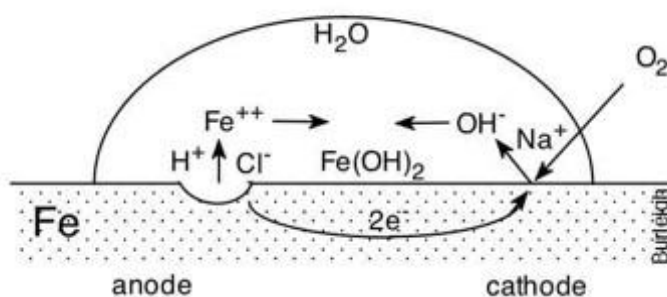
The overall reaction is being:



The ferrous ions are further oxidised by atmospheric oxygen to ferric ions which come out as rust in the form of hydrated ferric oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) and with further production of hydrogen ions.



Corrosion of Steel in a Waterdrop



Prevention of corrosion:

Corrosion can be prevented by the following methods:

1. **Barrier protection:** This is one of the simplest method corrosion (specially rusting of iron) in which a barrier is placed between metal surface and atmospheric air. This can be by the following methods:
 - (i) Coating the surface of metal with paint or some chemicals (e.g., bisphenol)
 - (ii) Protecting the surface by a thin film of oil or grease.
 - (iii) The metal is electroplated with metals like tin, nickel, zinc chromium, aluminium etc.
2. **Sacrificial protection:** In this method, the metal object is protected from corrosion by covering with a layer of metal (like Mg, Zn etc.) more reactive than the metal object which corrodes itself but saves the object. For example Zn is commonly for covering iron surfaces which is called galvanization.